

Experiment 5

Experiment 5.1

```
C exp5.1.c > ~
1 // WAP to read a list of integers and store it in a single dimensional array. Write a C program to print the second largest integer in a list of integers.
2 #include <stdio.h>
3
4 int main() {
5     int n, i;
6
7     printf("Enter the number of elements: ");
8     scanf("%d", &n);
9
10    int arr[n];
11
12    printf("Enter %d integers:\n", n);
13    for (i = 0; i < n; i++) {
14        scanf("%d", &arr[i]);
15    }
16
17    int largest, second_largest;
18
19    // Initialize largest and second largest
20    if (arr[0] > arr[1]) {
21        largest = arr[0];
22        second_largest = arr[1];
23    } else {
24        largest = arr[1];
25        second_largest = arr[0];
26    }
27
28    // Traverse the rest of the array
29    for (i = 2; i < n; i++) {
30        if (arr[i] > largest) {
31            second_largest = largest;
32            largest = arr[i];
33        } else if (arr[i] > second_largest && arr[i] != largest) {
34            second_largest = arr[i];
35        }
36    }
37
38    printf("The second largest integer is: %d\n", second_largest);
39
40    return 0;
41 }
42
```

Experiment 5.2

```
// WAP to read a list of integers and store it in a single dimensional array. Write a C program to count and display positive, negative, odd, and even numbers in an array
#include <stdio.h>

int main() {
    int n, i;
    int positive = 0, negative = 0, even = 0, odd = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++) {
        if (arr[i] > 0)
            positive++;
        else if (arr[i] < 0)
            negative++;

        if (arr[i] % 2 == 0)
            even++;
        else
            odd++;
    }

    printf("\nResults:\n");
    printf("Positive numbers: %d\n", positive);
    printf("Negative numbers: %d\n", negative);
    printf("Even numbers: %d\n", even);
    printf("Odd numbers: %d\n", odd);

    return 0;
}
```

Experiment 5.3

```
C exp5_3.c > ...
1 // WAP to read a list of integers and store it in a single dimensional array. Write a C
2 // program to find the frequency of a particular number in a list of integers.
3 #include <stdio.h>
4
5 int main() {
6     int n, i, num, count = 0;
7
8     printf("Enter the number of elements: ");
9     scanf("%d", &n);
10
11     int arr[n];
12
13     printf("Enter %d integers:\n", n);
14     for (i = 0; i < n; i++) {
15         scanf("%d", &arr[i]);
16     }
17
18     printf("Enter the number to find its frequency: ");
19     scanf("%d", &num);
20
21     for (i = 0; i < n; i++) {
22         if (arr[i] == num)
23             count++;
24     }
25
26     printf("The number %d appears %d time(s) in the array.\n", num, count);
27
28     return 0;
29 }
30
```

Experiment 5.4

```
C exp5_4.c > main()
1 // WAP that reads two matrices A (m x n) and B (p x q) and computes the product A
2 // and B. Read matrix A and matrix B in row major order respectively. Print both the
3 // input matrices and resultant matrix with suitable headings and output should be in
4 // matrix format only. Program must check the compatibility of orders of the matrices
5 // for multiplication. Report appropriate message in case of incompatibility.
6 #include <stdio.h>
7
8 int main() {
9     int m, n, p, q;
10    int i, j, k;
11
12    printf("Enter number of rows and columns of Matrix A (m n): ");
13    scanf("%d %d", &m, &n);
14
15    int A[m][n];
16
17
18    printf("Enter elements of Matrix A in row-major order:\n");
19    for (i = 0; i < m; i++) {
20        for (j = 0; j < n; j++) {
21            scanf("%d", &A[i][j]);
22        }
23    }
24
25    printf("Enter number of rows and columns of Matrix B (p q): ");
26    scanf("%d %d", &p, &q);
27
28    int B[p][q];
29
30    printf("Enter elements of Matrix B in row-major order:\n");
31    for (i = 0; i < p; i++) {
32        for (j = 0; j < q; j++) {
33            scanf("%d", &B[i][j]);
34        }
35    }
36
37    if (n != p) {
38        printf("Matrix multiplication not possible. Number of columns of A must equal number of rows of B.\n");
39        return 0;
40    }
41
42    int C[m][q]; // Resultant matrix
```

```
C exp5_4.c > main()
8  int main() {
43
44     // Initialize resultant matrix with 0
45     for (i = 0; i < m; i++) {
46         for (j = 0; j < q; j++) {
47             C[i][j] = 0;
48         }
49     }
50
51     // Compute product A * B
52     for (i = 0; i < m; i++) {
53         for (j = 0; j < q; j++) {
54             for (k = 0; k < n; k++) {
55                 C[i][j] += A[i][k] * B[k][j];
56             }
57         }
58     }
59
60     // Print Matrix A
61     printf("\nMatrix A:\n");
62     for (i = 0; i < m; i++) {
63         for (j = 0; j < n; j++) {
64             printf("%d\t", A[i][j]);
65         }
66         printf("\n");
67     }
68
69     // Print Matrix B
70     printf("\nMatrix B:\n");
71     for (i = 0; i < p; i++) {
72         for (j = 0; j < q; j++) {
73             printf("%d\t", B[i][j]);
74         }
75         printf("\n");
76     }
77
78     // Print Resultant Matrix C
79     printf("\nResultant Matrix (A * B):\n");
80     for (i = 0; i < m; i++) {
81         for (j = 0; j < q; j++) {
82             printf("%d\t", C[i][j]);
83         }
84         printf("\n");
85     }
86
87     return 0;
88 }
89
```